Hydrological Forecasting
Data, Modelling and Applications

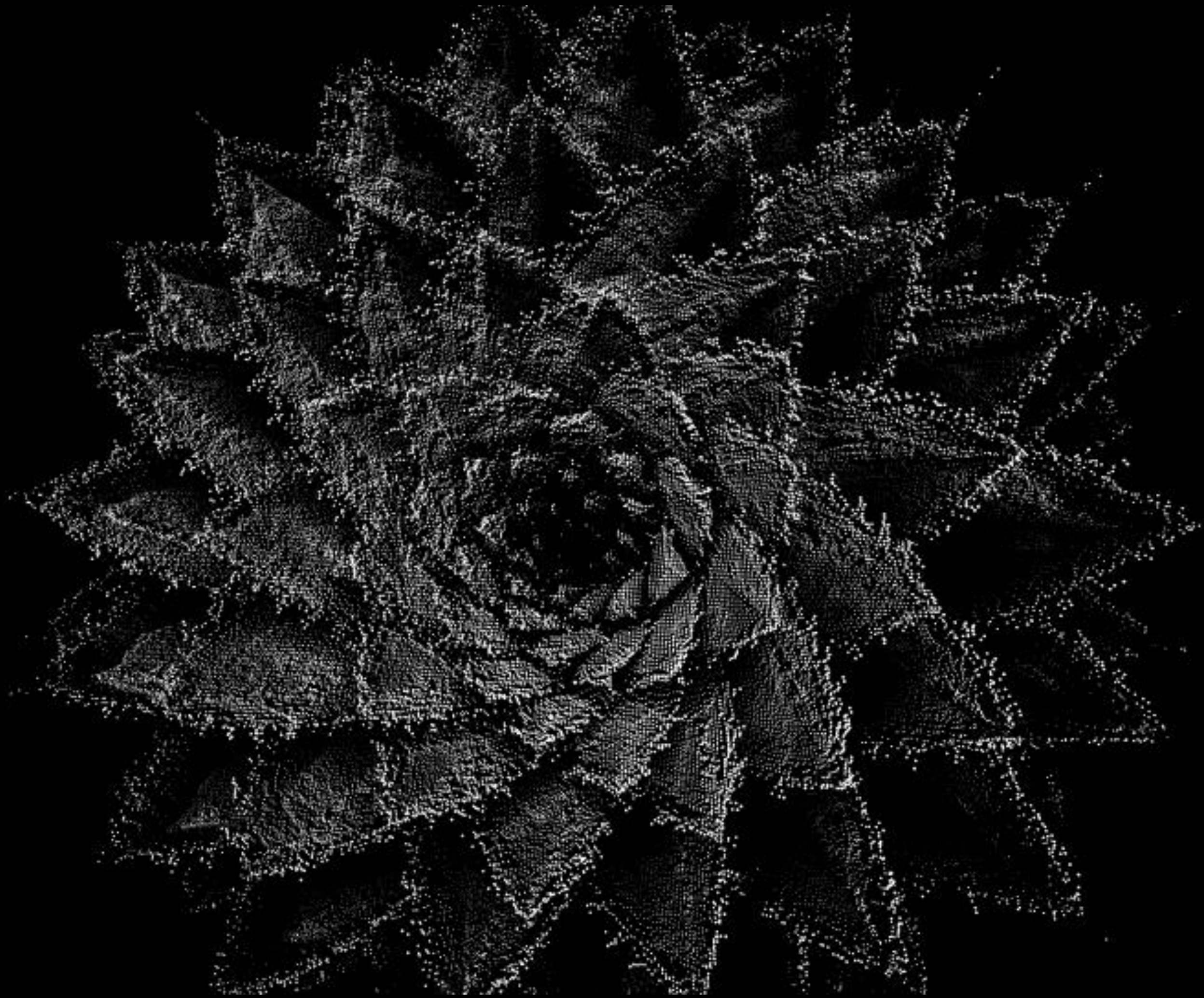
Training workshop
Accra, Ghana
27th-29th January 2026

NATIONAL CAPABILITY FOR GLOBAL CHALLENGES

UKCEH
Our planet. Decoded.

WASCAL
West African Science Service Centre on Climate Change and Adapted Land Use

The UK Centre for Ecology & Hydrology delivers science and impact across across **Air**, **Land** and **Water**, contributing to growth and resilience



UKCEH

UKCEH

Our research priorities

- Building resilience in a changing climate
- Enhancing ecosystem and human health
- Restoring biodiversity for a sustainable future

Where we are based

Our 700+ researchers deliver the data and insights that academics, governments and businesses need to solve urgent environmental challenges.

Edinburgh, Scotland

Lancaster, northern England

Bangor, north Wales

Wallingford, southern England

Accra, Ghana

Our history

UKCEH was formed in the 2000's through the merger of four UK research institutes dating back to 1960's.

We have a long history of **investigating**, **monitoring** and **modelling** environmental change.

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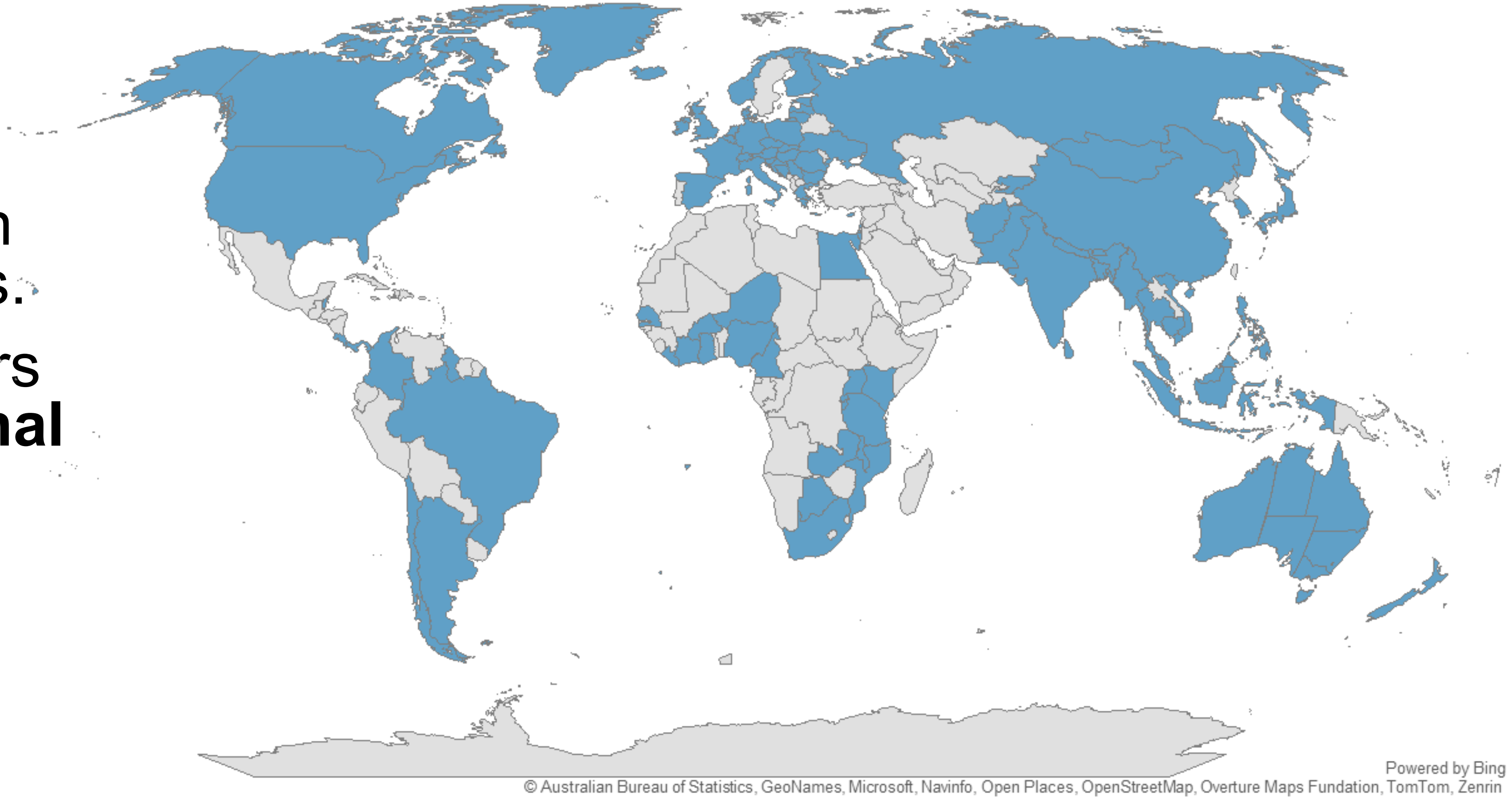
Our international work

- We collaborate with international partners based in over 90 countries.
- 67% of our papers **have international co-authors**
- We currently employ people form 44 different countries



[Welcome and introduction]

UKCEH water science

Water & Climate Science Area

Associate Science Director: Dr Glenn Watts

Informing action for water resources, flood management and water quality

Delivering observational networks and global scale services to inform water management

Water Resources & Drought Group

Head of Group: Jamie Hannaford

A focus on past, present and future water resources and drought

Developing operational tools and services to support water management

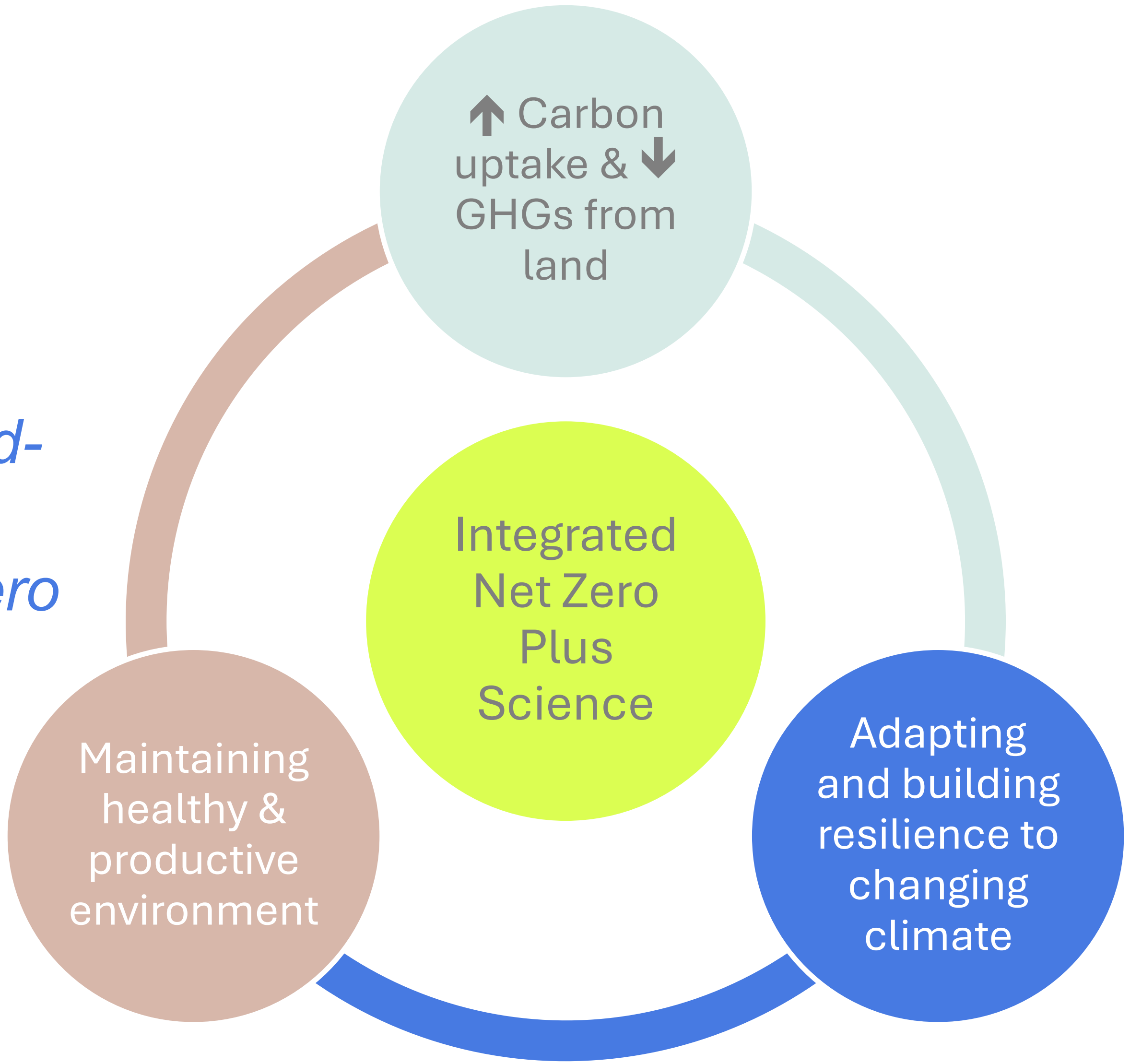
Hydrological data management and metadata

[Welcome and introduction]

UKCEH international research programme for global challenges

To enhance our understanding of the land-water-air system and its complex interactions that affect transition to Net Zero Plus and how we can build resilience to climate change

Work across three domains:
- SE Asia
- Sub-Saharan Africa
- Global



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[Welcome and introduction]

Building resilience to changing climate

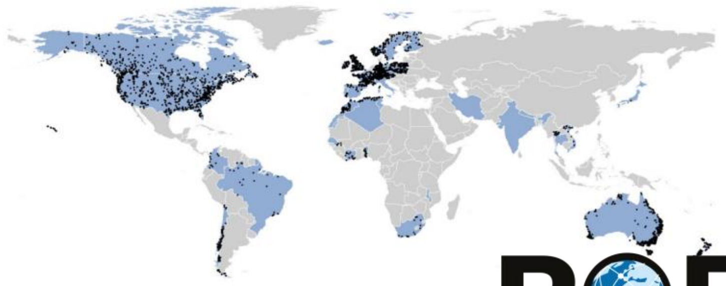


Fig. 1 Spatial distribution of countries participating in the ROBIN Net rk Ma ue din
Black dots are stations included in the first iteration of the ROBIN data

ROBIN
Reference observatory of basins for international hydrological climate change detection

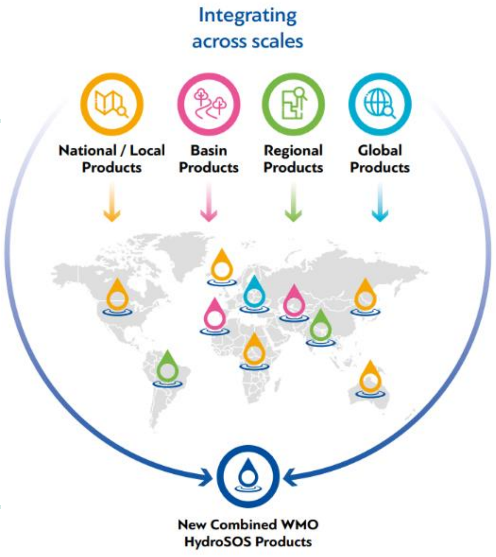
Aim

- Improved hydrological status and outlook methods and hydro-climate services to support climate change adaptation

Outputs

- New global model including water management
- New data products of global drought indicators and impacts
- Methods to blend global hydrological seasonal forecasts to improve skill
- Technical and implementation for WMO HydroSOS

WORLD METEOROLOGICAL ORGANIZATION
HydroSOS



Impact

- Enhanced resilience to natural hazards and adaptation to changing water availability
- Enhanced worldwide collaboration between researchers and hydromet agencies encourages sustainable transfer of science to operations

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[Welcome and introduction]

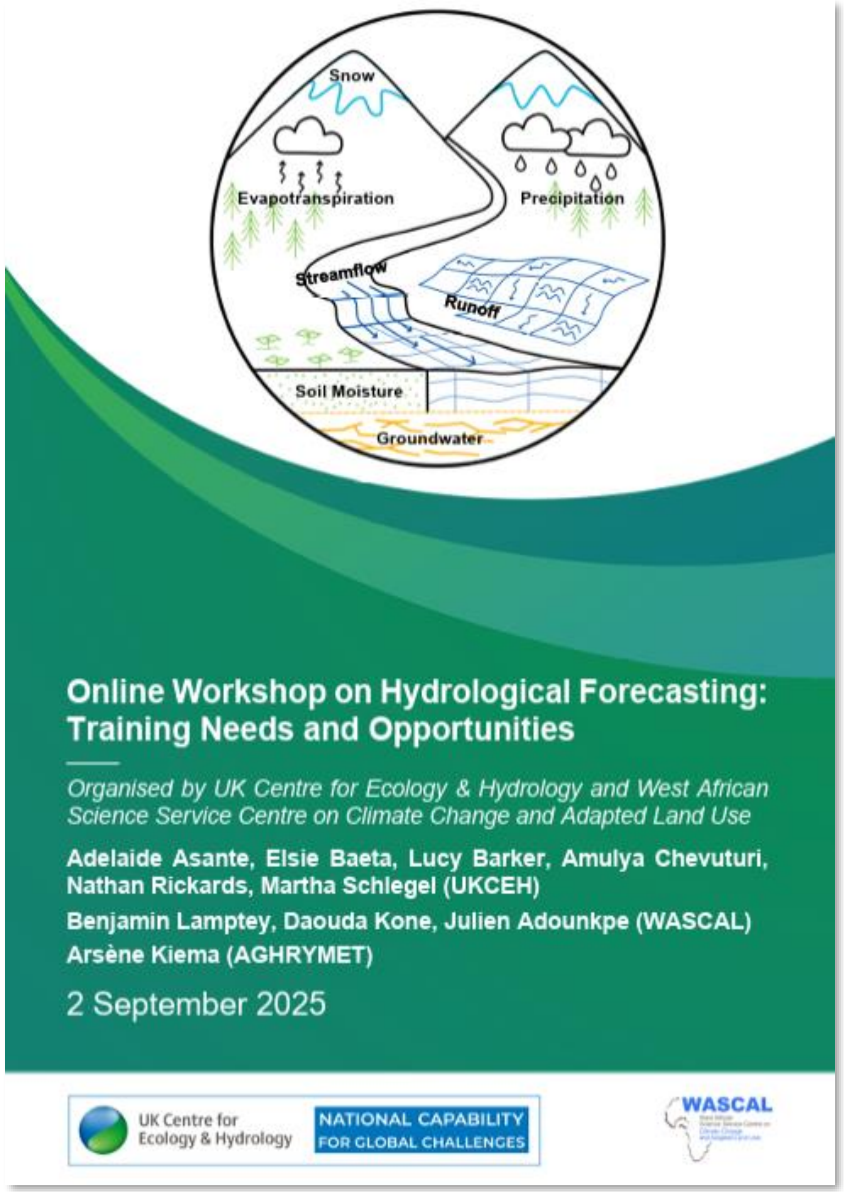
Pre-training webinar

To understand needs and opportunities for hydrological forecasting training

Interactive sessions on:

- Hydrological datasets
- Hydrological modelling and tools
- Hydrological seasonal forecast applications

Report link: https://nora.nerc.ac.uk/id/eprint/540414



Online Workshop on Hydrological Forecasting: Training Needs and Opportunities

Organised by UK Centre for Ecology & Hydrology and West African Science Service Centre on Climate Change and Adapted Land Use

Adelaide Asante, Elsie Baeta, Lucy Barker, Amulya Chevuturi, Nathan Rickards, Martha Schlegel (UKCEH)

Benjamin Lamptey, Daouda Kone, Julien Adounkpe (WASCAL)

Arsène Kiema (AGHRYMET)

2 September 2025

UK Centre for Ecology & Hydrology

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Training aim

Strengthen practical skills in hydrological forecasting, from data access and exploration to modelling and operational forecast applications

Hydrological Datasets

Water Resources Modelling

Seasonal Forecast Applications

[Welcome and introduction]

UKCEH training team

Lucy Barker
Senior Hydrological Analyst / NC-Intl. 3A Project Lead

Ezra Kitson
Environmental Data Scientist

Martha Schlegl
Scientific Project Manager

Amulya Chevuturi
Senior Hydroclimate Scientist / NC-Intl. 3A Task Lead

Eugene Magee
Hydrological Analyst

Katie Facer-Childs
Senior Hydrologist (supporting online)

Nathan Rickards
Hydrologist / NC-Intl. 3A Task Lead

Helen Baron
Hydrological Modeller

Adriana Calderon
Hydrological Analyst (supporting online)

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Roberto Silva Vara
Scientific Office
World Meteorological Organization, Geneva

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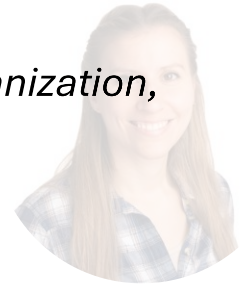

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Hydrological Modeller

Adriana Calderon
Hydrological Analyst (supporting online)

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[Welcome and introduction]

Overview of the week

Day 1: Amulya, Eugene
+ *Group photo!*

Day 2: Nathan, Helen
+ Group dinner

Day 3: Lucy, Eugene, Ezra

	Day 1: 27th January	Day 2: 28th January	Day 3: 29th January
	Theme: Hydrological Datasets	Theme: Water Resources Modelling	Theme: Seasonal Forecast Applications
08:30 - 09:00	**REGISTRATION**	---	---
09:00 - 10:00	Welcome & introductions	Talk: Reservoir modelling Talk: Introduction to Machine Learning (ML)	Talk: Introduction to forecast methods & applications
10:00 - 11:00	Hands-on: Computing setup and testing	Workshop: ML reservoir modelling 1	Workshop: Running a hydrological model
11:00 - 11:30	**BREAK**	**BREAK**	**BREAK**
11:30 - 13:00	Workshop: Remote data access	Workshop: ML reservoir modelling 2	Workshop: Running Ensemble Streamflow Prediction (ESP)
13:00 - 14:00	**LUNCH**	**LUNCH**	**LUNCH**
14:00 - 15:30	Workshop: Data analysis and visualisation	Talk: The benefits of a multi-reservoir ML model Workshop: ML reservoir modelling 3	Talk: WMO HydroSOS Workshop: Categorising a forecast product for enhanced decision making
15:30 - 16:00	**BREAK**	**BREAK**	**BREAK**
16:00 - 17:00	Q/A and wrap-up	Group discussion on workshop outcomes Q/A and wrap-up	Q/A, wrap-up and close
18:00	---	**GROUP DINNER**	---

Pre-training survey

[Welcome and introduction]

Introductions

Tour de table:

- Name
- Job title / position
- Organisation

Icon © Freepik, flaticon.com

Thank you

For more information
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@uk_ceh

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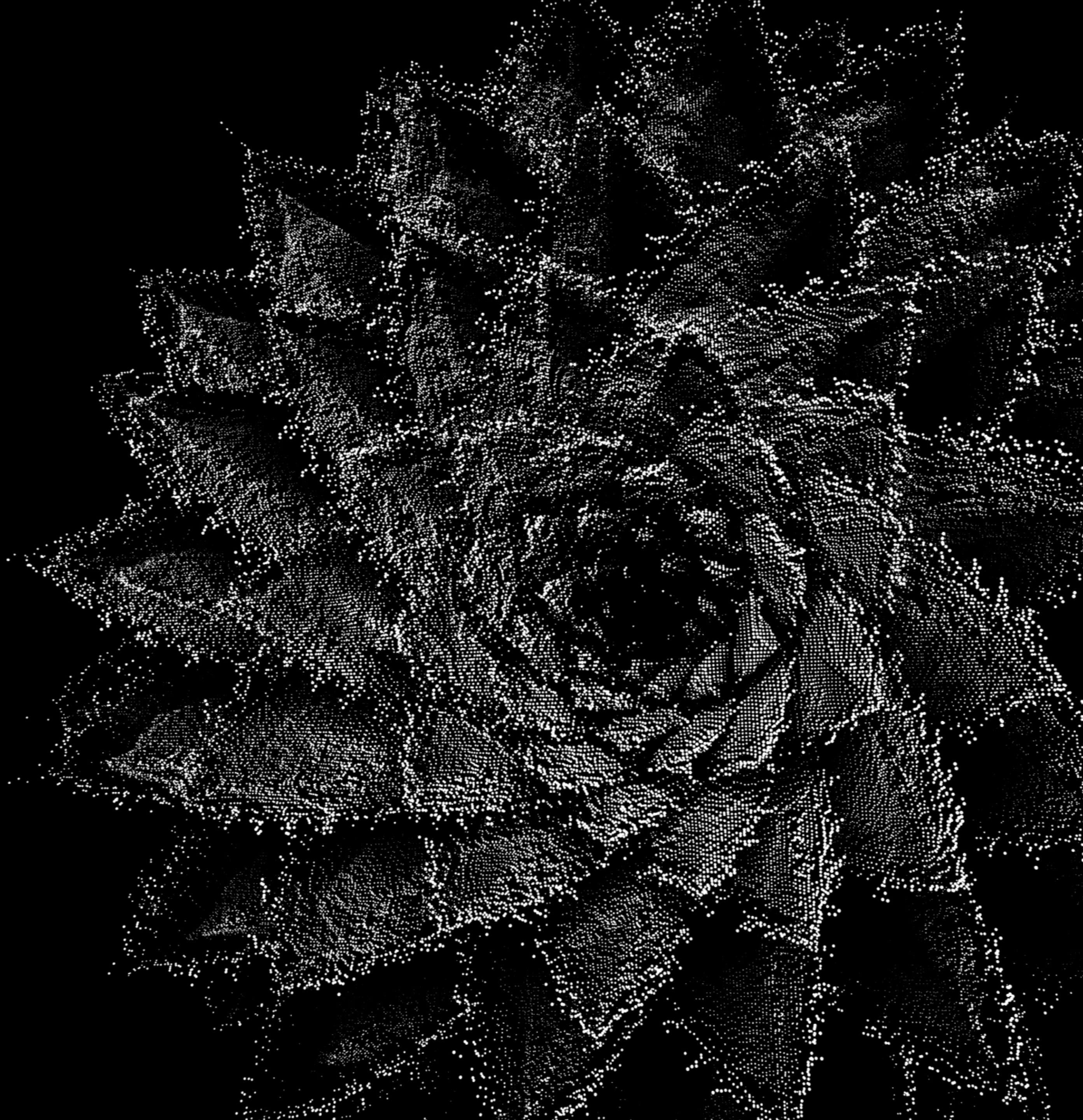

Adomi Bridge - Atimpoku

Computer set-up and testing

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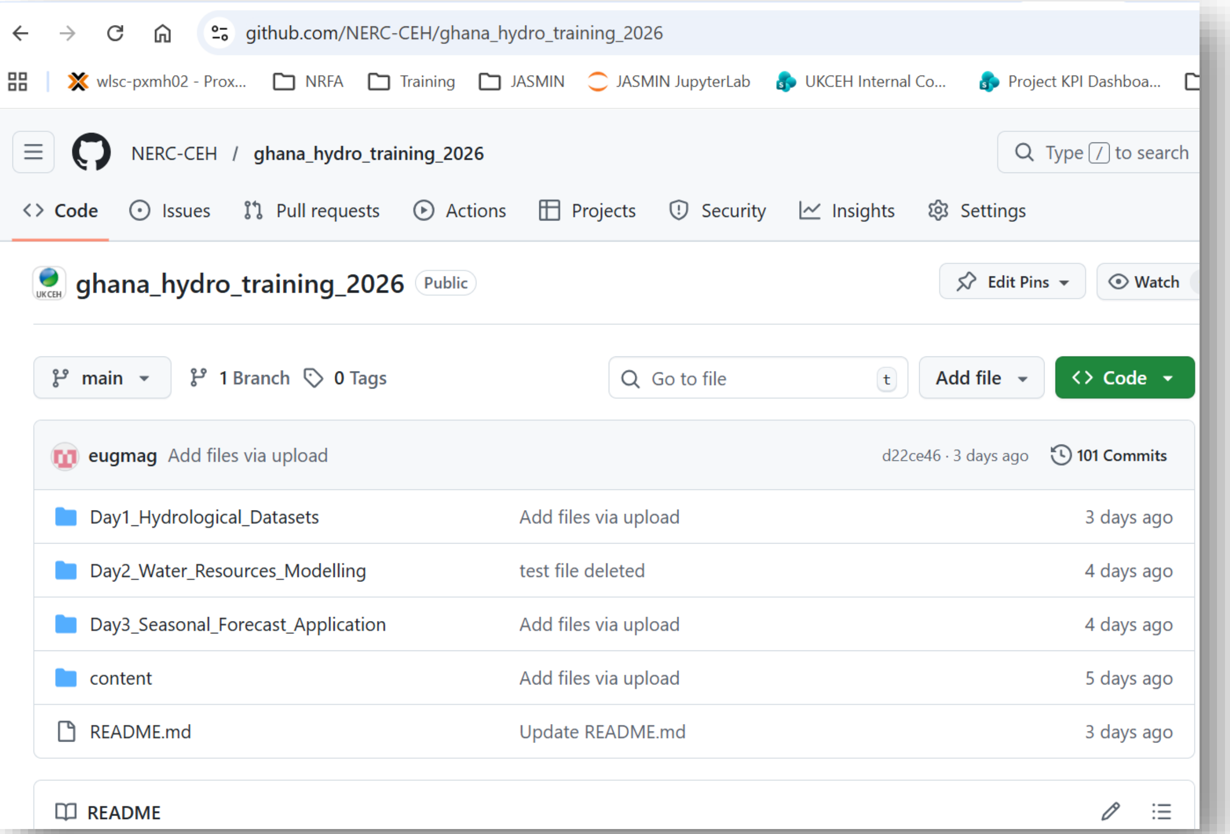
[Computing set-up and testing]

Google Colab + Github

- We will be running Jupyter notebooks on Google Colab stored in a Github repository
- Advantages of this approach:
  - You don't need to install Python or R on your laptop or need your own laptop processing
  - Collaborative, portable and free



colab.google

NERC-CEH/ ghana_hydro_training_2026

https://github.com/NERC-CEH/ghana_hydro_training_2026

1

- Sign into Google Colab with your Google account
- https://colab.google/

2

- Find the Training Github Repo
- **NERC-CEH/ghana_hydro_training_2026**

3

- Select the notebook you want (.ipynb) from the list.
- **Save a copy to your Drive** before editing and work from your own version.